Incident Analysis and Response Project

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Abstract

This report presents the implementation of an Incident Analysis and Response environment using SOC principles. The project focuses on deploying Wazuh agents, simulating Cyber Attacks such as brute force and DDoS, detecting these attacks, and applying active response mechanisms to block malicious activities. The main goal is to understand the full incident response lifecycle in a rea world simulated environment.

Introduction

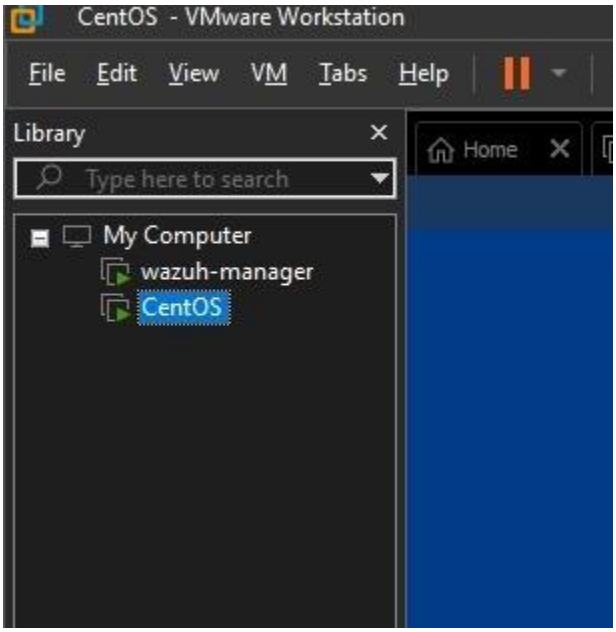
An Incident Analysis and Response system is responsible for detecting, analyzing, and responding to security incidents within an organization. With the rapid increase in cyber threats, organizations must adopt monitoring systems to ensure early detection and effective response.

This project aims to build a simulated SOC environment to analyze incidents and apply automated response techniques.

Step1: Environment Setup

The environment consists of two virtual machines:

- Cent OS Linux
- Ubuntu



Step2: Open Ubuntu and find the IP

```
wazuh@wazuh-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
 1000
    link/ether 00:0c:29:1e:07:bb brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.15.128/24 brd 192.168.15.255 scope global dynamic noprefixroute ens33
       valid_lft 1568sec preferred_lft 1568sec
    inet6 fe80::20c:29ff:fe1e:7bb/64 scope link
       valid_lft forever preferred_lft forever
```

Step3: Update and upgrade the Ubuntu system

```
wazuh@wazuh-VMware-Virtual-Platform:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for wazuh:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://eg.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://eg.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://eg.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
168 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  libllvm20
The following upgrades have been deferred due to phasing:
  initramfs-tools initramfs-tools-bin initramfs-tools-core
```

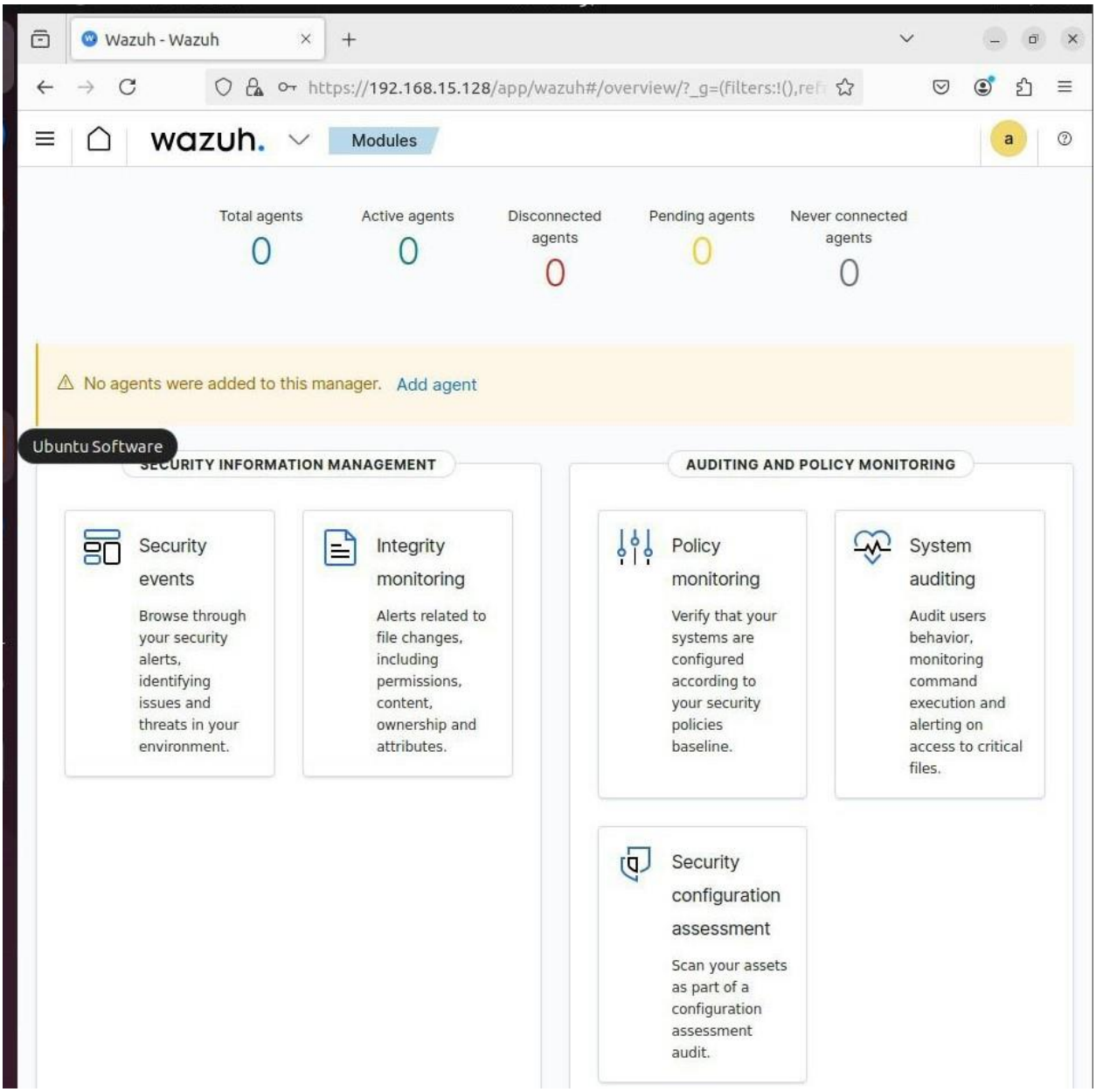
Step4: Install curl on ubuntu if it’s not already installed

```
azuh@wazuh-virtual-machine:~$ sudo apt install curl -y
eading package lists... Done
uilding dependency tree... Done
eading state information... Done
he following NEW packages will be installed:
 curl
 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
eed to get 194 kB of archives.
fter this operation, 455 kB of additional disk space will be used.
et:1 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.81.0-1ubuntu1.21 [194 kB]
etched 194 kB in 1s (214 kB/s)
electing previously unselected package curl.
(Reading database ... 203200 files and directories currently installed.)
reparing to unpack .../curl_7.81.0-1ubuntu1.21_amd64.deb ...
npacking curl (7.81.0-1ubuntu1.21) ...
etting up curl (7.81.0-1ubuntu1.21) ...
rocessing triggers for man-db (2.10.2-1) ...
```

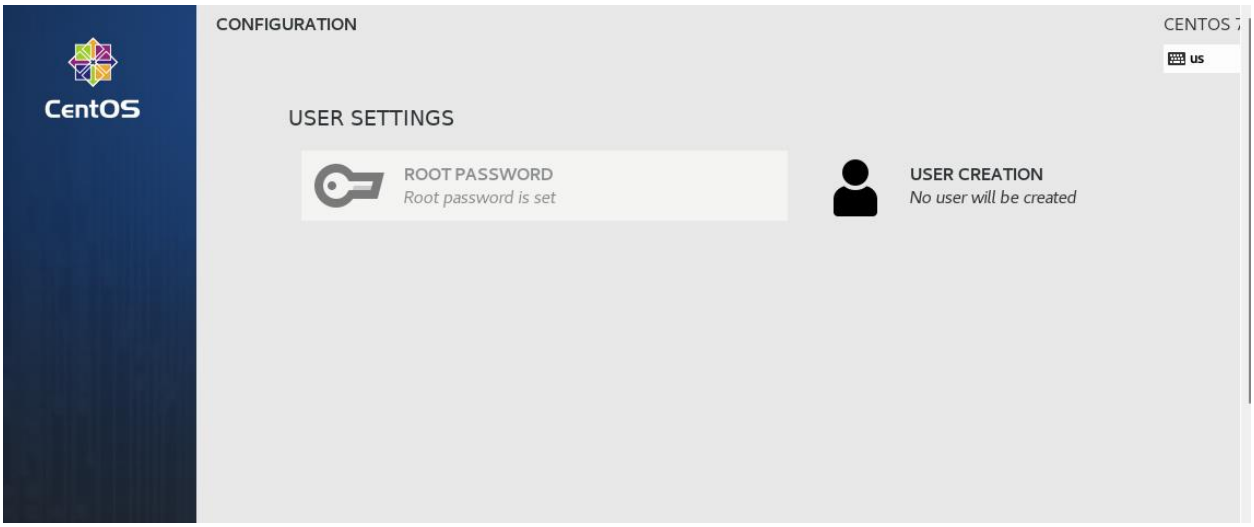
Step5: Install Wazuh

```
Terminal                                                    21:42 10 فبر
wazuh@wazuh-virtual-machine: ~
wazuh@wazuh-virtual-machine:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
wazuh@wazuh-virtual-machine:~$ ls
Desktop    Downloads  Pictures  snap       Videos
Documents  Music      Public    Templates  wazuh-install.sh
wazuh@wazuh-virtual-machine:~$ sudo bash wazuh-install.sh -a
10/02/2026 21:18:44 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
10/02/2026 21:18:44 INFO: Verbose logging redirected to /var/log/wazuh-install.log
10/02/2026 21:18:51 INFO: --- Dependencies ----
10/02/2026 21:18:51 INFO: Installing gawk.
10/02/2026 21:18:55 INFO: Wazuh web interface port will be 443.
10/02/2026 21:18:59 INFO: --- Dependencies ----
10/02/2026 21:18:59 INFO: Installing apt-transport-https.
10/02/2026 21:19:04 INFO: Wazuh repository added.
10/02/2026 21:19:04 INFO: --- Configuration files ---
10/02/2026 21:19:04 INFO: Generating configuration files.
10/02/2026 21:19:05 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
10/02/2026 21:19:06 INFO: --- Wazuh indexer ---
10/02/2026 21:19:06 INFO: Starting Wazuh indexer installation.
10/02/2026 21:23:01 INFO: Wazuh indexer installation finished.
10/02/2026 21:23:01 INFO: Wazuh indexer post-install configuration finished.
10/02/2026 21:23:01 INFO: Starting service wazuh-indexer.
10/02/2026 21:23:13 INFO: wazuh-indexer service started.
10/02/2026 21:23:13 INFO: Initializing Wazuh indexer cluster security settings.
10/02/2026 21:23:23 INFO: Wazuh indexer cluster initialized.
10/02/2026 21:23:23 INFO: --- Wazuh server ---
10/02/2026 21:23:23 INFO: Starting the Wazuh manager installation.
10/02/2026 21:25:32 INFO: Wazuh manager installation finished.
10/02/2026 21:25:32 INFO: Starting service wazuh-manager.
10/02/2026 21:25:47 INFO: wazuh-manager service started.
10/02/2026 21:25:47 INFO: Starting Filebeat installation.
10/02/2026 21:25:58 INFO: Filebeat installation finished.
10/02/2026 21:25:59 INFO: Filebeat post-install configuration finished.
10/02/2026 21:25:59 INFO: Starting service filebeat.
10/02/2026 21:26:00 INFO: filebeat service started.
10/02/2026 21:26:00 INFO: --- Wazuh dashboard ---
10/02/2026 21:26:00 INFO: Starting Wazuh dashboard installation.
10/02/2026 21:27:29 INFO: Wazuh dashboard installation finished.
10/02/2026 21:27:29 INFO: Wazuh dashboard post-install configuration finished.
10/02/2026 21:27:29 INFO: Starting service wazuh-dashboard.
10/02/2026 21:27:30 INFO: wazuh-dashboard service started.
10/02/2026 21:27:50 INFO: Initializing Wazuh dashboard web application.
10/02/2026 21:27:51 INFO: Wazuh dashboard web application initialized.
10/02/2026 21:27:51 INFO: --- Summary ---
10/02/2026 21:27:51 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: XX+h+*0iqfU9?RGeJVhZREofE5B?QR9F
10/02/2026 21:27:51 INFO: --- Dependencies ----
10/02/2026 21:27:51 INFO: Removing gawk.
10/02/2026 21:27:55 INFO: Installation finished.
```

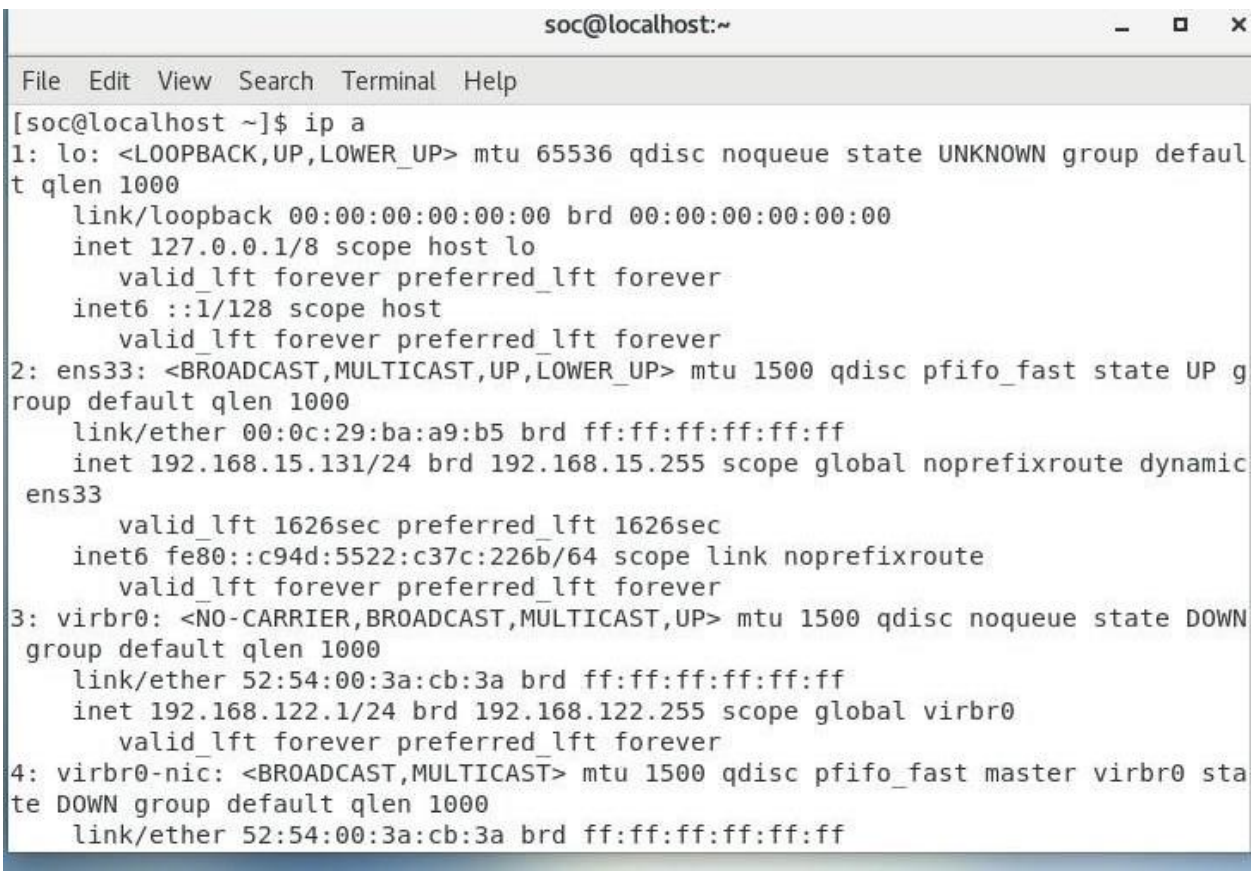
Step6: Open Wazuh



Step7: Open Cent OS and create user



Step8: Open Cent OS and find the IP address



Step9: Install nginx

```
omplete!
soc@localhost ~]$ sudo yum install nginx -y
oaded plugins: fastestmirror, langpacks
epository base is listed more than once in the configuration
epository updates is listed more than once in the configuration
epository extras is listed more than once in the configuration
oading mirror speeds from cached hostfile
pel/x86_64/metalink                                      | 3.6 kB     00:00
* epel: mirror.flo.c-f.ro
pel                                                      | 4.3 kB     00:00
1/3): epel/x86_64/updateinfo                               | 1.0 MB   00:00
2/3): epel/x86_64/group                                    | 399 kB   00:00
3/3): epel/x86_64/primary_db                               | 8.7 MB   00:03
esolving Dependencies
-> Running transaction check
--> Package nginx.x86_64 1:1.20.1-10.el7 will be installed
-> Processing Dependency: nginx-filesystem = 1:1.20.1-10.el7 for package: 1:nginx-1.20.1-10.el7.x86_64
-> Processing Dependency: libcrypto.so.1.1(OPENSSL_1_1_0)(64bit) for package: 1:nginx-1.20.1-10.el7.x86_64
-> Processing Dependency: libssl.so.1.1(OPENSSL_1_1_0)(64bit) for package: 1:nginx-1.20.1-10.el7.x86_64
-> Processing Dependency: libssl.so.1.1(OPENSSL_1_1_1)(64bit) for package: 1:nginx-1.20.1-10.el7.x86_64
-> Processing Dependency: nginx-filesystem for package: 1:nginx-1.20.1-10.el7.x86_64
-> Processing Dependency: libcrypto.so.1.1()(64bit) for package: 1:nginx-1.20.1-10.el7.x86_64
-> Processing Dependency: libssl.so.1.1()(64bit) for package: 1:nginx-1.20.1-10.el7.x86_64
-> Running transaction check
--> Package nginx-filesystem.noarch 1:1.20.1-10.el7 will be installed
--> Package openssl11-libs.x86_64 1:1.1.1k-7.el7 will be installed
-> Finished Dependency Resolution

ependencies Resolved
```

Step10: Start and enable nginx

```
[soc@localhost ~]$ sudo systemctl start nginx
[soc@localhost ~]$ sudo systemctl enable nginx
Created symlink from /etc/systemd/system/multi-user.target.wants/nginx.service to /usr/lib/systemd/system/nginx.se
vice.
[soc@localhost ~]$ curl -o wazuh-agent-4.7.5-1.x86_64.rpm https://packages.wazuh.com/4.x/yum/wazuh-agent-4.7.5-1.x
6_64.rpm && sudo WAZUH_MANAGER='192.168.15.128' rpm -ihv wazuh-agent-4.7.5-1.x86_64.rpm
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 9253k  100 9253k    0     0  3449k      0  0:00:02  0:00:02 --:--:-- 3450k
[sudo] password for soc:
warning: wazuh-agent-4.7.5-1.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID 29111145: NOKEY
Preparing...                          ################################# [100%]
Updating / installing...
   1:wazuh-agent-4.7.5-1              ################################# [100%]
[soc@localhost ~]$ sudo systemctl daemon-reload
[soc@localhost ~]$ sudo systemctl enable wazuh-agen
Failed to execute operation: No such file or directory
[soc@localhost ~]$ sudo systemctl enable wazuh-agent
Created symlink from /etc/systemd/system/multi-user.target.wants/wazuh-agent.service to /usr/lib/systemd/system/wa
uh-agent.service.
```

Step11: Linking the agent to Wazuh

Select the package to download and install on your system:

LINUX

- (●) RPM amd64
- ( ) RPM aarch64
- ( ) DEB amd64
- ( ) DEB aarch64

WINDOWS

- ( ) MSI 32/64 bits

macOS

- ( ) Intel
- ( ) Apple silicon

For additional systems and architectures, please check our documentation.

Server address:

Run the following commands to download and install the agent:

```
curl -o wazuh-agent-4.7.5-1.x86_64.rpm https://packages.wazuh.com/4.x/yum/wazuh-
agent-4.7.5-1.x86_64.rpm && sudo WAZUH_MANAGER='192.168.15.128' rpm -ihv wazuh-
agent-4.7.5-1.x86_64.rpm
```

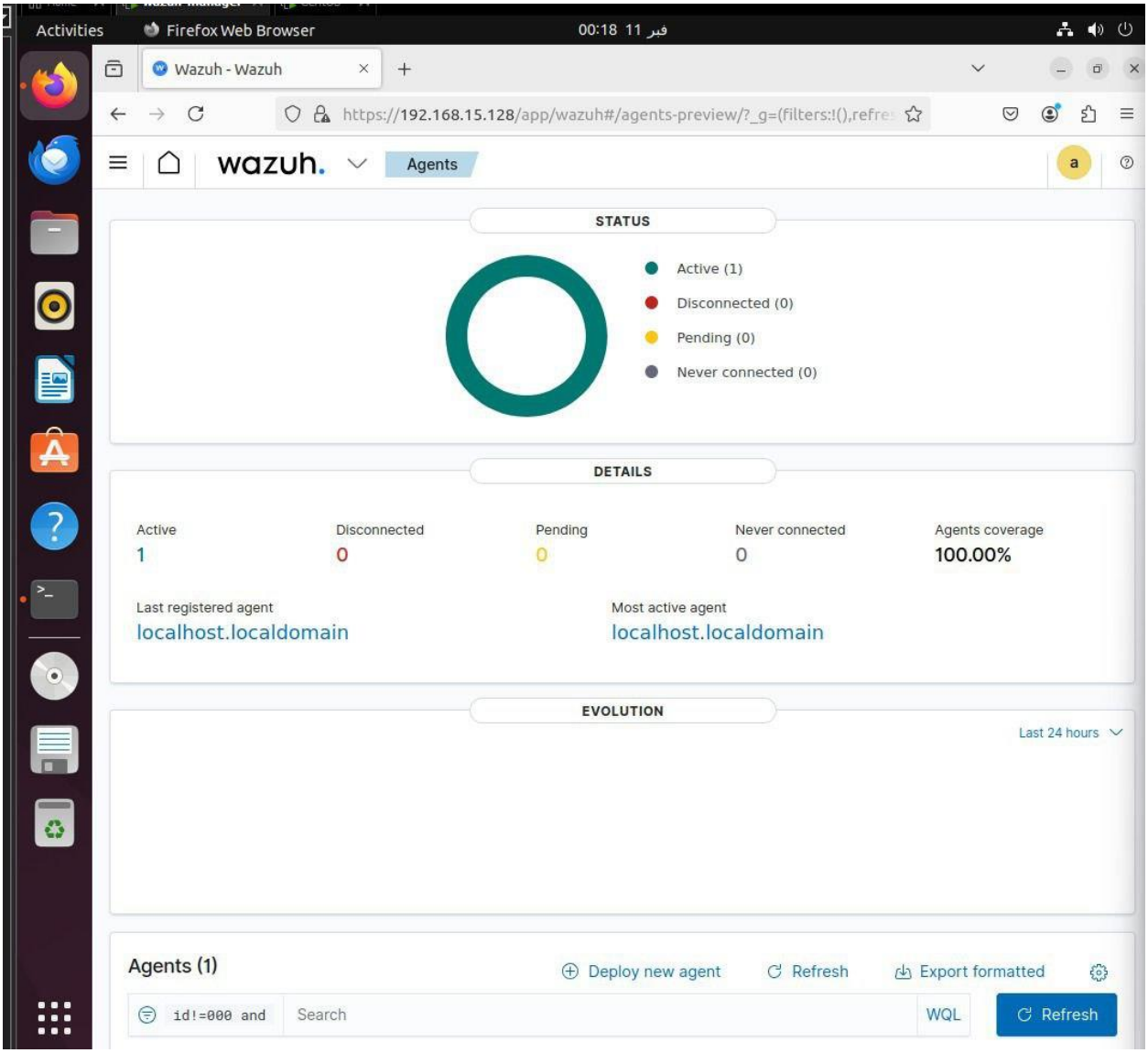
Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

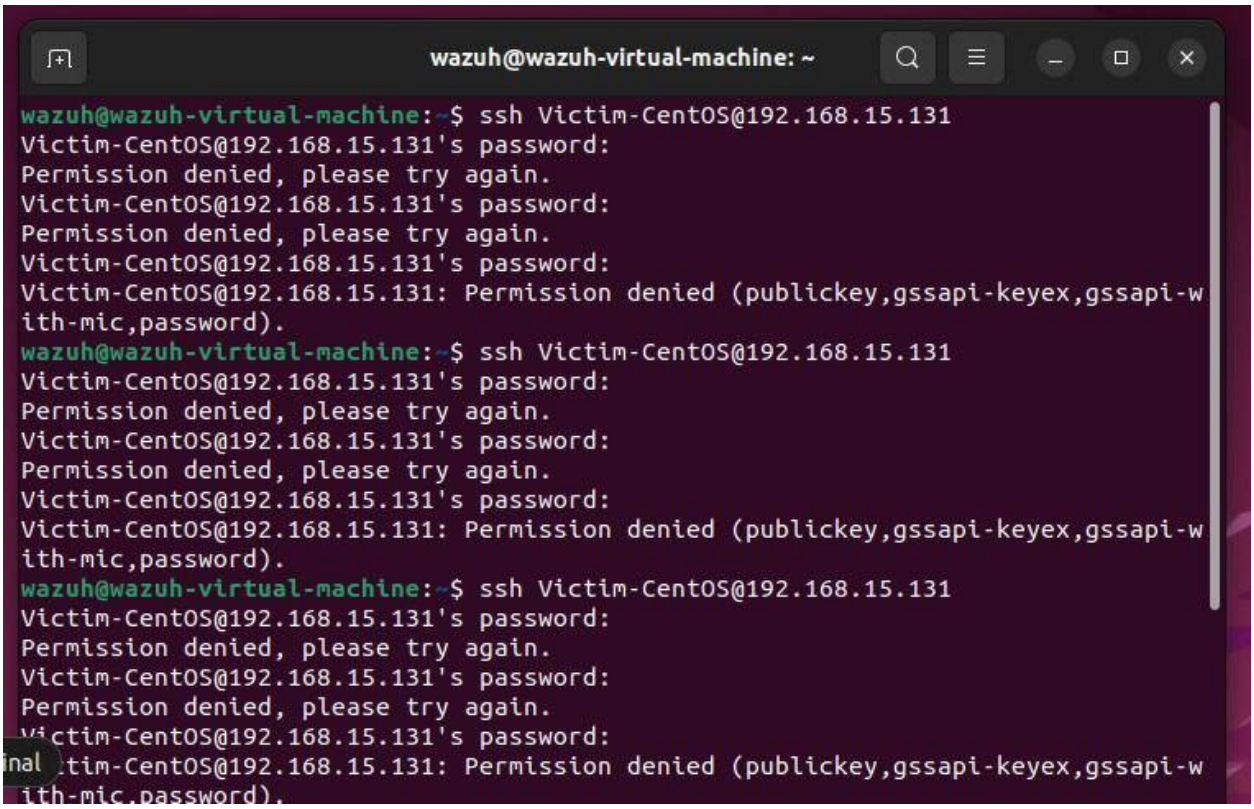
5 **Start the agent:**

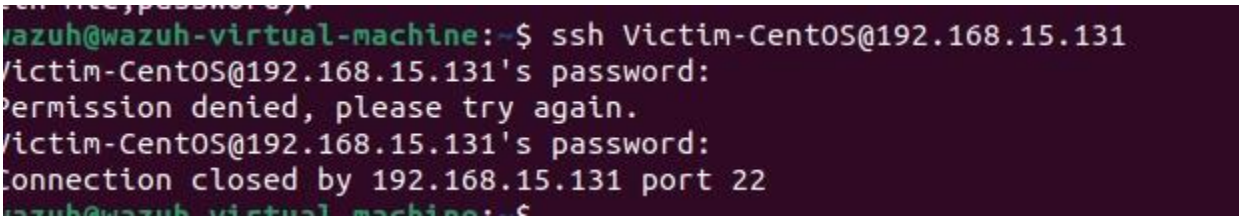
```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```


Activities
Firefox Web Browser
00:18 11 فبر
Wazuh - Wazuh
https://192.168.15.128/app/wazuh#/agents-preview/?_g=(filters:!(),refre
wazuh.
Agents
STATUS
Active (1)
Disconnected (0)
Pending (0)
Never connected (0)
DETAILS
Active
1
Disconnected
0
Pending
0
Never connected
0
Agents coverage
100.00%
Last registered agent
localhost.localdomain
Most active agent
localhost.localdomain
EVOLUTION
Last 24 hours
Agents (1)
Deploy new agent
Refresh
Export formatted
id!=000 and
Search
WQL
Refresh

Step12: Brute Force attack

A brute force attack is a trial-and-error hacking method that uses automated tools to try countless password, key, or credential combinations until the correct one is found, aiming to gain unauthorized access to accounts, systems, or encrypted data by sheer computational power rather than exploiting vulnerabilities.





Step13: Active Response

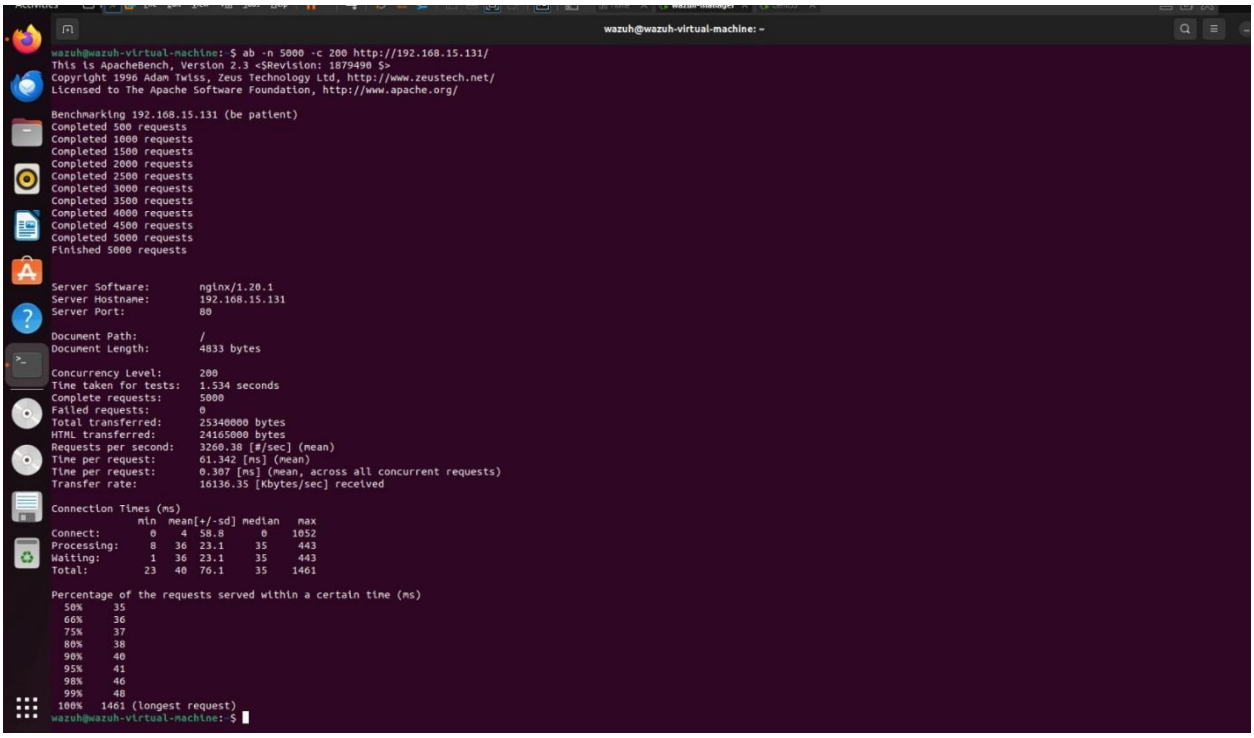
An attempt was made to implement the Active Response feature; however, it could not be executed successfully due to unexpected and unclear system errors.

Step14: DDOS Attack

A Distributed Denial-of-Service (DDoS) attack is a malicious attempt to disrupt normal traffic to a targeted server, service, or network by overwhelming it with a flood of Internet traffic from multiple compromised sources (a botnet).

```
wazuh@wazuh-virtual-machine: ~
wazuh@wazuh-virtual-machine:~$ sudo apt install apache2-utils -y
[sudo] password for wazuh:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  apache2-utils
Rhythmbox 1 newly installed, 0 to remove and 4 not upgraded.
Need to get 89.5 kB of archives.
After this operation, 409 kB of additional disk space will be used.
Get:1 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.18 [
89.5 kB]
Fetched 89.5 kB in 0s (239 kB/s)
Selecting previously unselected package apache2-utils.
(Reading database ... 324866 files and directories currently installed.)
Preparing to unpack .../apache2-utils_2.4.52-1ubuntu4.18_amd64.deb ...
Unpacking apache2-utils (2.4.52-1ubuntu4.18) ...
Setting up apache2-utils (2.4.52-1ubuntu4.18) ...
Processing triggers for man-db (2.10.2-1) ...
wazuh@wazuh-virtual-machine:~$
```

```
[root@localhost ~]# sudo tail -f /var/log/nginx/access.log
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
192.168.15.128 - - [12/Feb/2026:09:48:49 -0800] "GET / HTTP/1.0" 200 4833 "-" "ApacheBench/2.3" "-"
```



References

[1] https://youtu.be/Qr_yUzmxYeI?si=nQvVc2PrZQlG2Cmw

[2] https://youtu.be/JTGMWH2w2p4?si=33wG-ASM_I3jtYqR

[3] https://youtu.be/BTR2JSNgld4?si=NRLxNdRVIHA-P2uT

[4] https://youtu.be/VGoKrnguEtY?si=SWExGXdyjx6kPQlz

[5] https://youtu.be/pXBk2upn8Wo?si=T4ld4Y20gYS_6soF

[6] https://youtu.be/1cO81wmebZo?si=CwDNXmlLEhHoRsdB